App Flow

How a person uses CoolRoute, and what the system does at each step

CoolRoute: Pedestrian Routing with Shade and Heat-Exposure Budgets

1. The user journey in seven steps

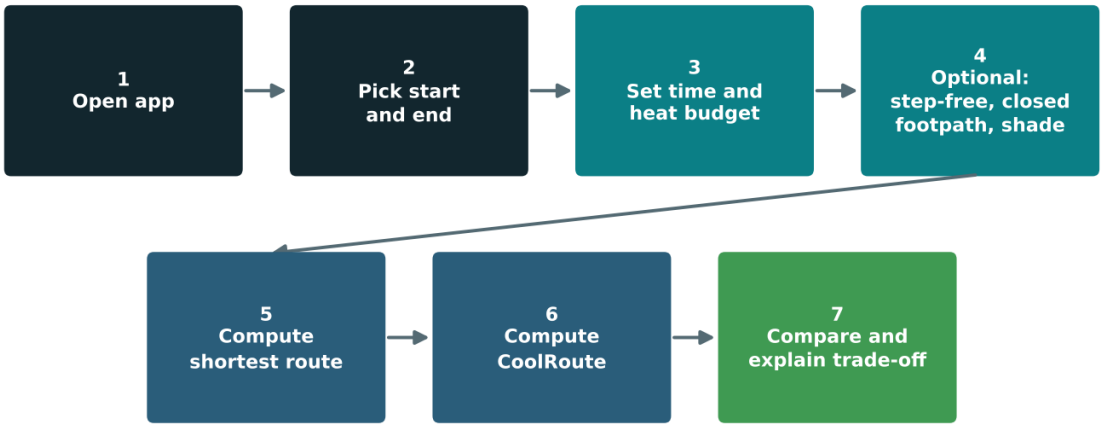


Figure 1. From opening the app to an explained route choice.

Step	What the user does	What the app does
1	Opens the app	Shows the map with a sample trip and both routes already drawn, so the first screen is never empty.
2	Taps a junction to set A and B	Snaps the tap to the nearest junction and redraws both routes.
3	Moves the time slider and the heat-budget slider	Recomputes sun strength and both routes instantly (the search takes milliseconds on a small map).
4	Optional: turns on Step-free, turns off Shade, or closes a footpath	Removes blocked segments before searching. Step-free is a hard rule.
5	Looks at the shortest route	Shows distance, time, heat, shade share, effort and data confidence.
6	Looks at CoolRoute	Shows the same numbers next to the shortest route, plus the change in each.
7	Reads the verdict	One sentence: how much longer, how much cooler, and why.

2. What happens inside one request

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User changes a control
  -> read state: A, B, hour, budget, step_free, shade_on, closed footpaths
  -> sun   = sun(hour)                        # 0 at night, about 0.77 at 1 pm
  -> graph = remove closed segments; if step_free: remove non-step-free segments
  -> base  = shortest path (lambda = 0)
  -> if no path: show 'No route found' message and stop
  -> budget = base.heat * budget_pct / 100
  -> if base.heat <= budget: cool = base
     else: search lambda by bisection (40 steps) so that cool.heat <= budget
  -> build warnings: budget not met, large detour, missing data, step-free note
  -> draw routes, fill table, write the verdict sentence
```

3. Edge cases the app must handle

Situation	What the user sees	Why
No route exists (closed footpaths or step-free leave no path)	'No route found' with a hint to reopen a footpath or turn off step-free	Better than drawing a wrong route
Heat budget too strict for this map	Coolest route available plus warning 'budget cannot be met'	The route is still useful; the user must know the limit
Cooler route is more than 20% longer	Warning 'Large detour' and a hint to raise the budget	The problem statement asks to show when a cooler route adds too much distance
Some streets have missing shade data	Dashed streets, a count of missing segments, 'approximate' note	Honest about uncertainty
Night time (no sun)	Both routes are the same; message explains why	Heat is zero, so there is nothing to optimise
Shortest route already fits the budget	'CoolRoute keeps it'	No pointless detour
User sets A equal to B	Tap on the same junction is ignored	Avoids a zero-length trip

4. Screen layout

- **Phone:** map first, then scenario controls, then results, then the sun chart. One column, no sideways scrolling.
- **Laptop (for the demo):** map on the left, controls and results on the right, so judges see everything without scrolling.
- Every tap target is at least 32 px high. Colours work in light and dark mode.

5. Accessibility rules built into the flow

- Step-free is a switch, not a hidden setting.
- Routes differ by **line style and colour**, not colour alone (thick halo versus thin solid line).
- Text explanation always accompanies the map, so screen-reader users get the same information.
- Next step for production: audio and large-text versions of the verdict.